RAMAZAN FARUK OĞUZ

Senior Software Engineer · Java · Apache Spark · Kafka · Azure



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PROFILE

Senior software engineer with 7+ years across mainframe, fintech and automotive. At Critical TechWorks (BMW Group) I build large-scale Java and Apache Spark pipelines on Azure that carry more than 5 million order records per day. MS in Computer Science (GPA 3.86), with published research on automated UI testing for interaction-heavy web systems.

EXPERIENCE

Rockstar Developer (Senior Software Engineer track)

Jul 2022 – Present

Critical TechWorks (BMW Group) · Porto, Portugal

- **AI agents in the pipeline.** I helped bring an orchestrator into our pipeline that runs AI agents from GitHub Actions, starting with pull request review to take the repetitive parts off reviewers' hands, and I drive company-wide knowledge sharing on it through workshops.
- As a main contributor, I helped migrate BMW Group's mobility reporting from batch-only pipelines to Java + Apache Spark Structured Streaming on Azure HDInsight, now handling more than 5 million order records per day from Kafka.
- I design the pipelines for idempotent writes, explicit schema evolution and checkpointed state, so replays and retries stay safe for downstream consumers at that volume.
- I maintain the batch Spark jobs that recalculate aggregates for consumers who need reconciled numbers, keeping batch and streaming views consistent.
- REST services expose curated pipeline outputs, so downstream teams consume data without coupling to the Spark layer.
- Cut pipeline runtime by 50% by profiling bottlenecks and reworking partitioning, join strategy and skew handling.
- I review pull requests, mentor junior engineers, and work directly with data engineers, product owners and BMW stakeholders in Portugal, Germany and Hungary to turn requirements into pipeline designs.

Java · Apache Spark · Structured Streaming · Kafka · Azure HDInsight · GitHub Actions · LLM APIs

Software Engineering Consultant

Jul 2022 – Aug 2023

Aubay Portugal · Porto, Portugal

- Delivered BMW Group's mobility data platform as an external consultant, embedded with the Critical TechWorks team on Java/Spark pipelines while owning production work end to end.
- Worked on the same pipelines as the permanent engineers, which led to a direct Critical TechWorks contract in August 2023.

Java · Apache Spark · Docker

Senior Software Engineer

Apr 2021 – Jul 2022

Garanti BBVA Teknoloji · Istanbul, Turkey

- Built and maintained the core credit-card cycle services (statement, billing, interest calculation) in Java 8 / Spring Boot, treating correctness as the design constraint rather than a testing phase.
- Owned a Spring Boot service exposing cycle data through REST and gRPC contracts, keeping consumer systems decoupled from internal model changes.
- Implemented card-number generation within BIN ranges, including check-digit derivation and product assignment.
- Worked with Kafka and Apache NiFi streaming flows and designed for replay and idempotency, so reprocessing could not corrupt a balance.
- Held the quality bar with JUnit 4 + Mockito coverage, SonarQube gates and code review; shipped through Jenkins, JFrog Artifactory, Bitbucket and OpenShift.
- Supported production incidents and mentored junior engineers.

Java 8 · Spring Boot · REST · gRPC · Kafka · Apache NiFi · Jenkins · OpenShift

Junior Software Engineer

Jul 2020 – Mar 2021

Garanti BBVA Teknoloji · Istanbul, Turkey

- Joined the credit-card core cycle without a banking background, learning the domain rules behind statements and billing before changing code.
- Implemented modules in Java 8 / Spring Boot and used JUnit/Mockito tests to trace defects from a wrong figure back to its source data.

Java 8 · Spring Boot · JUnit · Mockito

Software Expert Mar 2020 – Jun 2020

Smartera · Istanbul, Türkiye

- Helped build an exploratory data analysis tool that let non-technical users interrogate and visualize their own datasets.
- Worked across preprocessing and visualization in Python (pandas, NumPy, Matplotlib), turning messy exports into something decision-makers could read.

Python · pandas · NumPy · Matplotlib

Student Assistant Oct 2019 – Apr 2020

Yıldız Technical University · Istanbul, Türkiye

- Standardized lab PC images with Symantec Ghost and migrated student and senior projects onto a shared platform, so faculty and students could find them.

Scholarship Holder Oct 2019 – Mar 2020

TÜBİTAK BİLGEM · Kocaeli, Türkiye

- Continued the A* path-finding project with a focus on visualization and integration, building the visualization layer and joining the components into one working flow so the algorithm could be inspected and reused, not only read in code.

Summer Intern Jun 2019 – Aug 2019

TÜBİTAK BİLGEM · Kocaeli, Türkiye

- Implemented and deployed A* in native Python, applied to maze path-finding and structured so the same approach could extend to autonomous driving scenarios.

Summer Intern Jun 2018 – Aug 2018

Yıldız Technical University Software Quality Lab · Istanbul, Turkey

- Ran a structured SQL vs NoSQL comparison on the same workload, using MySQL and MongoDB with the benchmark program written in C#.
- Published the results as a conference paper at Akademik Bilişim, under the supervision of Dr. Mehmet S. Aktaş.

EDUCATION

Yıldız Technical University — MS, Computer Science 2020 – 2022

GPA 3.86 / 4.00 · Thesis: End-to-end automated UI testing framework for web applications with intensive user-system interaction.

Yıldız Technical University — BSc, Computer Engineering 2016 – 2020

GPA 3.75 / 4.00 · Graduated 3rd in class.

SKILLS

Languages	Java, Python, C#, SQL, COBOL, PL/I, JCL
Big Data	Apache Spark (Structured Streaming, batch), Apache Kafka, Apache NiFi, Azure HDInsight
Backend	Spring Boot, REST, gRPC, Microservices
Cloud & DevOps	Microsoft Azure, Docker, OpenShift, Jenkins, JFrog Artifactory, Bitbucket, Git
Data Stores	IBM Db2, MySQL, MongoDB, z/OS
Testing	JUnit 4, Mockito, SonarQube

PUBLICATIONS

1. Y. Uygun, R.F. Oğuz, E. Ölmezoğulları, M.S. Aktaş. "On the large-scale graph data processing for user interface testing in big data science projects." *IEEE BigData 2020*, pp. 2049–2056.
2. I. Erdem, R.F. Oğuz, E. Ölmezoğulları, M.S. Aktaş. "Test script generation based on hidden Markov models learning from user browsing behaviors." *IEEE BigData 2021*, pp. 2998–3005.
3. R.F. Oğuz, M. Öz, E. Ölmezoğulları, M.S. Aktaş. "Extracting information from large scale graph data: case study on automated UI testing." *Euro-Par 2021 Workshops, LNCS*, pp. 364–375.
4. R.F. Oğuz, I. Erdem, E. Ölmezoğulları, M.S. Aktaş. "End-to-end automated UI testing workflow for web sites with intensive user–system interactions." *International Journal of Software Engineering and Knowledge Engineering*, 2022.
5. R.F. Oğuz, Y. Uygun, M.S. Aktaş, İ. Aykurt. "On the use of technical analysis indicators for stock market price movement direction prediction." *SIU 2019*.

132 citations, h-index 6 — scholar.google.com/citations?user=JLtGY48AAAAJ

LANGUAGES

English — professional working proficiency · Portuguese — intermediate · German — beginner